

MCQs

1. Which of the following is a mutable data type in Python?

- a) tuple
- b) string
- c) list
- d) int

ANS: c) list

2. What is the output of `print(type(10/2))`?

- a) int
- b) float
- c) double
- d) number

ANS: b) float

3. Which keyword is used to define a function in Python?

- a) func
- b) def
- c) function
- d) define

ANS: b) def

4. PIP stands for?

- a) Python Interpreter Program
- b) Pip Installs Packages
- c) Program Installation Package
- d) Python Installation Program

ANS: b) Pip Installs Packages

5. Which command saves changes to local git repo?

- a) git push
- b) git commit
- c) git add
- d) git update

ANS: b) git commit

Short Questions

1. What are the differences between scripting and programming?

Ans. Programming produces a complex standalone application and is usually done in a compiled language, such as Java. Scripting is just the automation of an existing environment and is typically done in an interpreted language, such as Python. All scripting is programming, but not vice-versa.

2. Explain the evolution of Python.

Ans. Python was Created in 1991 by Guido van Rossumut. It's updated versions 2.0 and 3.0 was created in 2008. It went from simple scripting to one of the dominant languages in modern web development, data science, and even AI.

3. What is Git and why is it used?

Ans. Git is a free, open-source version control system that tracks changes in source code. It is used to allow multiple developers to collaborate effectively and manage project history safely.

4. What is the purpose of a virtual environment?

Ans. Through the use of a virtual environment, developers can have an isolated place to manage all the dependencies of their project and, in turn, keep projects from conflicting by ensuring that the libraries and versions are separated from all other projects.

5. Explain advantages of Jupyter Notebook.

Ans. Jupyter is a platform that brings together code, graphics and text in one document making it easier for users to conduct experiments with data interactively and share those findings with others. This makes Jupyter particularly well-suited for people working in the field of data science.